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SARDAR PATEL UNIVERSITY
BCA (I – Semester) Examination
US01BCA53: Fundamental of Computer Organization
DATE: 14/02/2022, Monday
TIME: 03:00 P.M TO 05:00 P.M

Q:1 Multiple choice questions.

Maximum Marks: 70

[10]

1. Which parts of the computer perform arithmetic calculations?
A. ALU B. Registers C. Logic bus D. Input device
2. The ALU and CU jointly known as :
A. RAM B. ROM C. CPU D. None of the above
3. Storage of 1 KB means the following number of bytes _____.
A. 1000 B. 964 C. 1024 D. 1064
4. Extra bit added to a string of bits to detect errors is known as _____.
A. Additional bit B. Correction bit C. Parity bit D. updation bit
5. In _____ parity the total number of 1's in the complete code including parity bit is even number.
A. Even B. Odd C. Even-odd D. None
6. The maximum number stored in 8 bits using 1's complement is _____.
A. 127 B. 128 C. 129 D. -127
7. Multiprocessor is referred as _____.
A. SISD B. SIMD C. MISD D. MIMD
8. Which memory is permanent type memory?
A. ROM B. RAM C. EPROM D. EEPROM
9. Which one is output device?
A. keyboard B. Scanner C. mouse D. plotter
10. Which printer gives the best result of printing?
A. Dot-matrix printer B. Laser printer C. Inkjet printer D. None of these

Q:2 Fill in the Blanks and True/False.

[08]

1. ASCII equivalent of B is _____.
2. In pipeline a single instruction is divided in _____ stages.
3. _____ is a pointing device.
4. ALU stands for _____.
5. 1 MB=2048 bytes. [True/False]
6. The full form of EPROM is Effective PROM. [True/False]
7. Dot matrix printer is an example of Serial printer. [True/False]
8. The mapping of characters onto integers is called a Character Code. [True/False]

[1]

[P.T.O.]

[20]

Q:3 Answer the following questions. (Any ten)

1. Differentiate – Hardware and Software.
2. Explain second generation of computer.
3. Perform following Conversions: 1) Decimal to Binary – 1020
2) Binary to Decimal – 110011
4. Explain 1's complement method with example.
5. Explain ASCII character code in brief with example.
6. Define Data Path. List Types of Instructions in CPU Organization.
7. What do you mean by ROM?
8. What is Instruction register?
9. Explain Flash drive.
10. What is the function of Scanner?
11. Differentiate Impact Printer and nonimpact printer.
12. List types of output devices. Also write down use of each device.

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Q:4 Answer the following questions. (Any Four)

1. Explain Keyboard with all kind of keys.
2. List type of printers and explain any one type of printer in detail.
3. Explain array processor with diagram.
4. What is Primary Memory? List different types of Primary memory and explain in detail.
5. Explain Hamming code method with example.
6. Perform following binary addition and subtraction examples.
1) $1101 + 1110$
2) $00110 + 011110$
3) $101000 - 010101$
4) $11110 - 000011$
7. Explain hexa-decimal number system with example.
8. List methods for Representation of Integers. Explain any one in detail.

[2]